Simple Linear Regression

Linear Regression

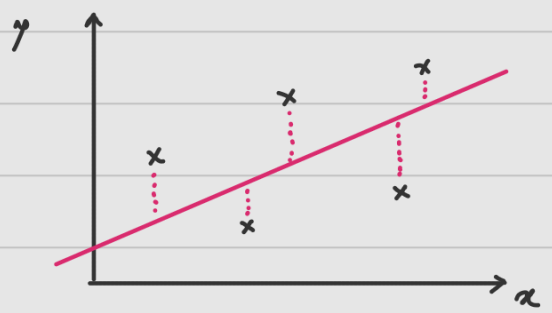
- Closed form sol$^{\underline{n}}$ → Ordinary Least Square → Suaitable for lower dimensions
- Non-closed form sol$^{\underline{n}}$ → Gradient Descent → Sutaible for higher dimensions

Ordinary Least Squares

$y = mx + b$ where $b = \bar{y} - m\bar{x}$

$$m = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

Derivation



$$\text{Error} = \sum (y_i - \hat{y}_i)^2$$
$$= \sum (y_i - (mx_i + b))^2$$

now, we need to minimize it.

$$E(m,b) = \sum (y_i - (mx_i + b))^2$$

Note, y is squarely related to 'b'

Let us first optimize it for b,

Using minima / maxima concepts, let

$$\frac{\partial E}{\partial b} = 0$$

i.e. $$\frac{\partial}{\partial b} \Sigma (y_i - mx_i - b)^2 = 0$$

$$\rightarrow \Sigma \frac{\partial}{\partial b} (y_i - mx_i - b)^2 = 0$$

$$\rightarrow \Sigma\, 2(y_i - mx_i - b) \cdot (-1) = 0$$

$$\rightarrow -2\, \Sigma (y_i - mx_i - b) = 0$$

$$\rightarrow \Sigma y_i - \Sigma m x_i - nb = 0$$, dividing both sides with n

$$\rightarrow \frac{\Sigma y_i}{n} - \frac{\Sigma m x_i}{n} - \frac{nb}{n} = 0$$

$$\rightarrow \bar{y} - m\bar{x} - b = 0$$

$$\rightarrow b = \bar{y} - m\bar{x}$$

Now, substituting value of b in E, we get

$$E = \Sigma (y_i - mx_i - \bar{y} + m\bar{x})^2$$

$$\frac{\partial}{\partial m} \Sigma (y_i - mx_i - \bar{y} + m\bar{x})^2 = 0$$

$$\rightarrow -2\, \Sigma (y_i - mx_i - \bar{y} + m\bar{x})(x_i - \bar{x}) = 0$$

$$\rightarrow \Sigma \{(y_i - \bar{y}) - m(x_i - \bar{x})\}(x_i - \bar{x}) = 0$$

$$\rightarrow \sum \{ (y_i - \bar{y})(x_i - \bar{x}) - m(x_i - \bar{x})^2 \} = 0$$

$$\rightarrow \sum (y_i - \bar{y})(x_i - \bar{x}) - \sum (x_i - \bar{x})^2 m = 0$$

$$\rightarrow \boxed{m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2}}$$

Multiple Linear Regression

Generalized Linear Regression Format

- Assuming N samples for training
- # Features = M

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_N \end{bmatrix}_{N \times 1} = \begin{bmatrix} 1 & x_{1,1} & x_{1,2} & \dots & x_{1,M} \\ 1 & x_{2,1} & x_{2,2} & \dots & x_{2,M} \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 1 & x_{N,1} & x_{N,2} & \dots & x_{N,M} \end{bmatrix}_{N \times (M+1)} \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_M \end{bmatrix}_{(M+1) \times 1}$$

$$\boxed{\hat{Y} = X\theta}$$

Loss $L = e^T e = (y-\hat{y})^T (y-\hat{y})$

Now, $\hat{y} = X\theta$

$$L = (y - X\theta)^T (y - X\theta)$$

$$= (y^T - (X\theta)^T)(y - X\theta)$$

Same term

$$= y^T y - (X\theta)^T y - X\theta y^T + (X\theta)^T (X\theta)$$

$$= y^T y - 2 y^T X\theta + \theta^T X^T X \theta$$

Now, we need to find the slope wrt θ

$$\frac{\partial L}{\partial \theta} = \frac{\partial}{\partial \theta}\left(y^T y - 2y^T X\theta + \theta^T X^T X \theta \right) = 0$$

$$\rightarrow 0 - 2y^T X + \frac{\partial}{\partial \theta} \theta^T X^T X \theta$$

$$\frac{\partial}{\partial z} z^T A z = 2 z^T A$$

here $A = X^T X$

$z = \theta$

$$\rightarrow -2y^T X + 2\theta^T X^T X$$

$$\rightarrow \cancel{2}\theta^T X^T X = \cancel{2} y^T X$$

$$\rightarrow \theta^T = y^T X (X^T X)^{-1}$$

$$\rightarrow (\theta^T)^T = \left(y^T X (X^T X)^{-1} \right)^T$$

$$\rightarrow \theta = \{(X^T X)^{-1}\}^T \{ y^T X \}^T$$

$$\rightarrow \theta = (X^T X)^{-1} (X^T y)$$

Gradient Descent

$$b_{new} = b_{old} + \alpha \text{ Slope}_{b = b_{old}}$$

$$m_{new} = m_{old} + \alpha \text{ Slope}_{m = m_{old}}$$

Loss $= \Sigma(y_i - \hat{y})^2$

slope wrt. b can be given as $\frac{\partial L}{\partial b}$

so, $\frac{\partial L}{\partial b} = \frac{\partial}{\partial b} \Sigma(y_i - \hat{y})^2 = \frac{\partial}{\partial b} \Sigma(y_i - (mx_i + b))^2$

$$= -2 \Sigma(y_i - mx_i - b)$$

slope wrt. b at $b = t$ can be given as $-2 \Sigma(y_i - mx_i - t)$

thus the value of b updates as

$$b_{new} = b_{old} + \alpha \left(-2 \Sigma(y_i - mx_i - b_{old})\right)$$

slope wrt. m can be given as $\frac{\partial L}{\partial m}$

so, $\frac{\partial L}{\partial m} = \frac{\partial}{\partial b} \Sigma(y_i - \hat{y})^2 = \frac{\partial}{\partial m} \Sigma(y_i - (mx_i + b))^2$

$$= 2 \Sigma(y_i - mx_i - b)(-x_i)$$

slope wrt. m at $m = t$ can be given as $-2 \Sigma(y_i - tx_i - b) x_i$

thus the value of m gets updated as

$$m_{new} = m_{old} + \alpha \left(-2 \Sigma(y_i - m_{old} x_i - b) x_i\right)$$

Note,

In gradient descent, update happens after seeing all the points. this version of gradient descent is called batch gradient descent.

A version of gradient descent where update happens after seeing each point is called stocastic gradient descent.

A version of gradient descent where update happens after seeing k points is called mini-batch gradient descent.

Linear Regression with gradient descent

$\hat{y} = x\theta$

$\epsilon = y - \hat{y}$

squared error $= \epsilon^T \epsilon = (y-\hat{y})^T (y-\hat{y})$

$$\rightarrow \epsilon^T \epsilon = (y - x\theta)^T (y - x\theta)$$
$$= y^T y - \theta^T x^T y - x\theta y^T + \theta^T x^T x \theta$$
$$= y^T y - 2x\theta^T x^T y + \theta^T x^T x \theta$$

$$\frac{\partial}{\partial \theta} \epsilon^T \epsilon = 0 - 2x^T y + 2\theta^T x^T x$$
$$= 2x^T (x\theta - y)$$

$$\theta_{new} = \theta_{old} - \alpha \{2x^T (x\theta - y)\}$$